

IMPERIAL

Instabilities in Quasi-3d

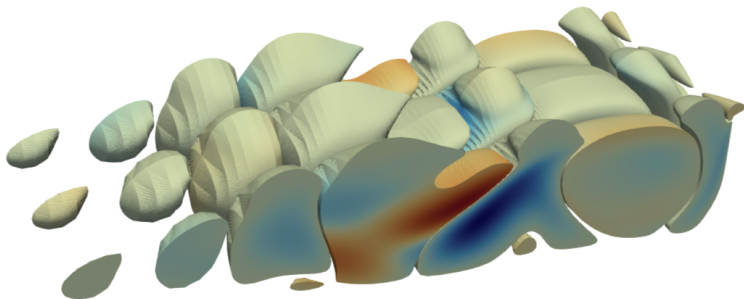
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quasi 3d incNS

- I tried $L_z = 2d, 3d, 4d$ so far, where d is the gap depth in the gap with $d = 3$ and $w = 25$ (just before the Hopf bifurcation in $2d$).

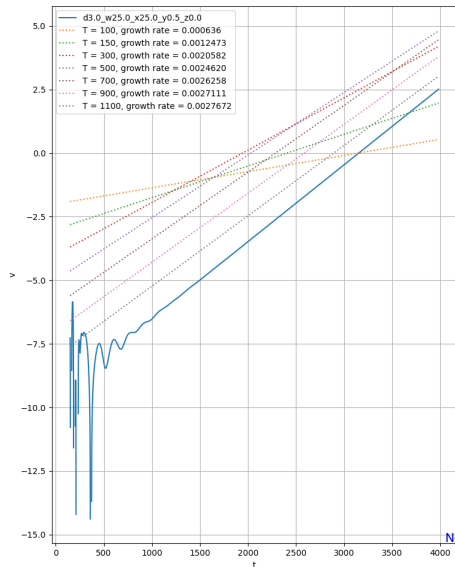
Quasi 3d with $L_z = 2d$ (and also $L_z = 3d$, although I need to double check)

- Unstable complex mode:
- $\sigma = 0.0047955, \omega = 0.0010237$
- v component profile:



Quasi 3d with $L_z = 4d$

- The non convergence of the eigenmodes to the same growth rate as we change the integration time between arnoldi iterations is because I think the structure is very long (so more time is needed).
- $\sigma \sim 0.003$ (less than the cavity mode)
- v component profile:



Marcello Paper 2d stability curve

Ma = 0.5 (Marcello paper):

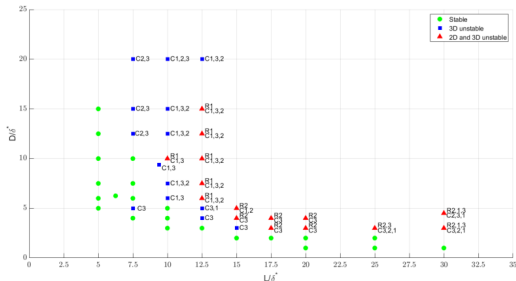
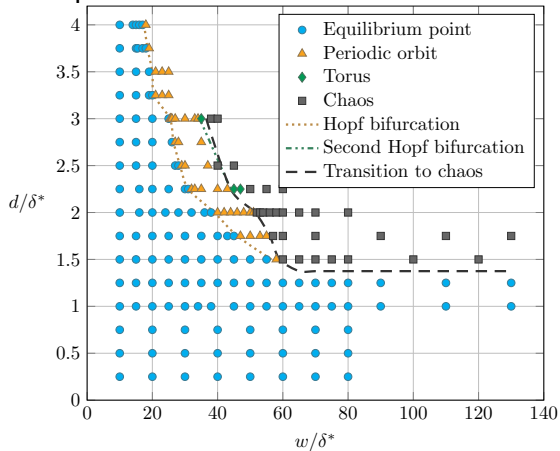


Fig. 2 Map of modes observed by LST

Incompressible case:



$d = 3, w = 10$ ($Ma = 0.6$)



$d = 1.5, w = 30$ ($Ma = 0.6$)



$d = 1.25, w = 60$ ($Ma = 0.6$)

